Experiment 2: Material response under mechanical load

Objectives

Understand material responses (brittle and ductile) under loading, learn techniques to measure stress and strain, and identify various material properties from a simple uniaxial test.

Background

The relationship between stress and strain for a material can be established using its experimental response at a macroscopic scale under different loading conditions. A tensile test is performed on a relatively slender member by pulling in the direction of its axis. Each material exhibits a unique stress-strain response under tensile loading. While a ductile material undergoes significant plastic deformation before fracture, a brittle material exhibits little deformation before fracturing.

Before conducting the experiment, we must first answer the following questions:

1.) Precisely define strength, stiffness, toughness, ductility, hardness, brittleness, flexure strength, and strain energy considering the material response.
2.) What is the working principle of a Load Cell and what are the types of load cells?
3.) How can we measure the strain – and types of devices used for measuring strains?
4.) Why are there pre-defined geometries (ASTM standards) for testing out materials under compression (typically cylindrical) and tension (dog-bone samples)?

Some online resources

You may watch the following resources to refresh your memory on torsion of circular shafts.

UTM tensile testing (9 minutes)

MTS UTM Tensile Testing (3 mins) (This is the machine in our lab here)

In-depth mechanical testing training (Recommended - long video but provides complete understanding of an Instron machine - 2hrs)

Each material is different - and so is their uniaxial material response (10 mins)

Metal Testing: Metal - Rectangular Dogbone Coupon Testing

Polymer Testing: Polymer - Rectangular Dogbone Coupon Testing

Above are official videos from Instron. Observe the difference in the stress strain curves.

The experiment

Task 1:

The universal testing machine is used to load a tensile test specimen and obtain the stress vs strain response. Your job is to set up specimens made of different materials (aluminium, mild steel, polymer) in the UTM and perform the tensile test until complete failure. A load cell, which is a transducer, measures the force applied to the test specimen. Assuming that there are no extensometers or any other strain measuring devices in the laboratory – estimate the strains using a video of the test captured on your mobile. Take help from the lab staff and TAs if you need to.

Task 2:

Write a report on the entire exercise. The report must not exceed 3 pages. You can tailor your report as per the following proposed structure, but you are free to deviate from it if you wish to.

1.) INTRODUCTION: State your objectives, and the answers to questions 1-3 stated in the "Background" section above.
2.) PROCEDURE: Describe the details of the procedure used to conduct the experiment.
3.) OBSERVATIONS: Report all the experimental data recorded. Describe your observations, insights, and any calculations you perform. Use plots and tables to compare experiments with theory. Does your experimental data agree with the theoretical predictions? If not, explain the possible reasons.
4.) CONCLUSIONS: Describe what you understood from this entire exercise.